

SMART INDIA HACKATHON 2026



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HACKATHON
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TITLE PAGE

- **Problem Statement ID – SIH26104**
- **Problem Statement Title- AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks**
- **Theme-Miscellaneous**
- **PS Category- Software**
- **Team ID- 140037**
- **Team Name (Registered on portal)- HackSpire 2..0**

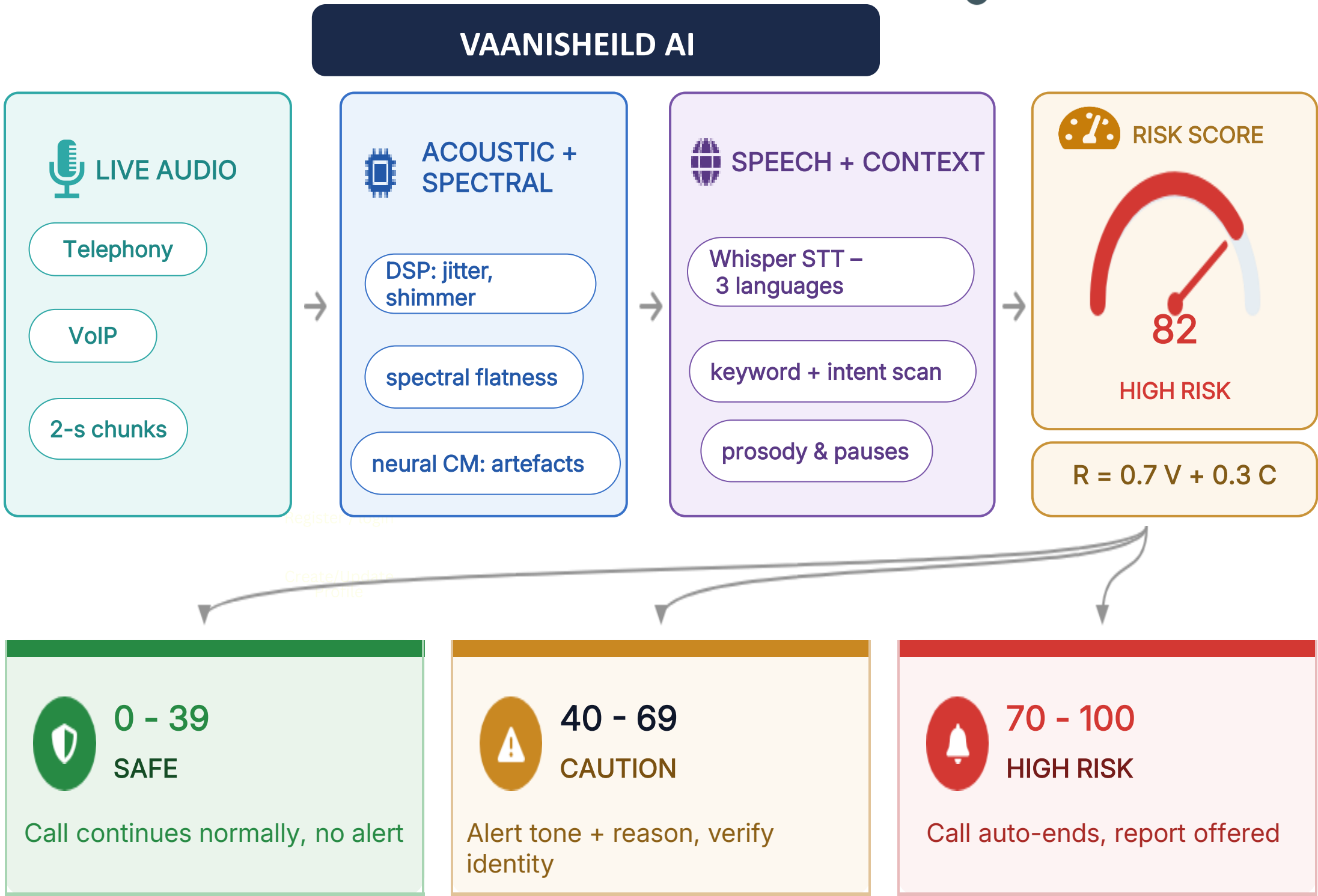


PROBLEM OVERVIEW & EXPECTED SOLUTION

- **Problem:** AI-cloned voices bypass traditional call verification, enabling real-time fraud via impersonation calls.
- **Solution:** Real-time voice-authenticity engine that scores impersonation risk live and alerts before sensitive actions occur.

PROPOSED SOLUTION

- **Background Call Guardian :** Live floating risk badge on the call screen - Truecaller-style, no manual upload.
- **Graduated Live Response :** Green (Safe) continues , Yellow beeps + warns , Red (High-Risk) auto-ends the call.
- **Real-Time Voice Deepfake Detection:** DSP + AI models analyze 2s audio chunks for cloned/spoofed voice signatures.
- **Multilingual & Accent-Aware:** Hindi, English, Marathi support for Indian users and regional accents.
- **Context-Aware Risk Fusion:** Combines voice authenticity with NLP keyword risk (OTP, transfer, urgent, digital arrest).
- **Explainable & Privacy-First:** “Why Flagged?” reasons on a dashboard; minimal retention, edge/on-device option.



TECH STACK

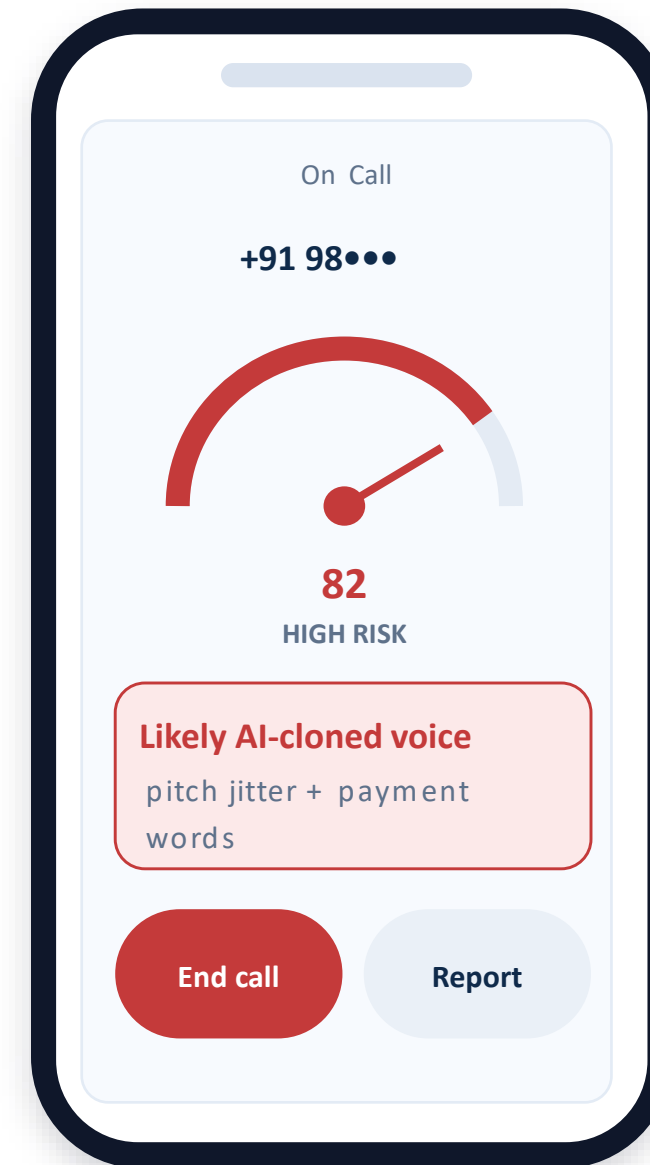
- › **Mobile App:** Java / Kotlin
- › **Backend:** FastAPI, Python, Uvicorn
- › **Voice / Speech AI:** Librosa (DSP), Whisper (multilingual STT), AASIST/RawNet2
- › **Data & Storage:** PostgreSQL, Redis, SQLite
- › **Infra:** Docker, Render, Vercel, Edge/TFLite
- › **Security:** AES-256, JWT/OAuth2, DPDP-aligned, audit logs

COMPOSITE RISK FORMULA

$$\text{Risk} = 0.7 \times \text{VoiceScore} + 0.3 \times \text{ContextScore}$$

VoiceScore = f(pitch jitter, shimmer, spectral flatness, HF-energy ratio) per 2s chunk

ContextScore = min(100, 25 * matched risk-keywords in live transcript)



WHAT YOU GET

- **Warning before the ask**
risk moves past 40 mid-sentence
- ✓ **Explainable, never a black box**
reason codes on every score
- ✗ **No Audio Retention**
features only, discarded at call end
- ⚡ **Runs on the edge**
~2 s re-scoring, no cloud round-trip

 **Multilingual**
• Hindi • English
• Marathi

 **Explainable**
“Why Flagged?”

 **Privacy-first edge**
processing

 **B2B API + consumer**
app

 **Live Prototype Link :** [click here](#)

 **Youtube Link :** [click here](#)

FEASIBILITY

→ Operational Feasibility

Easily integrates with existing **bank & telecom** systems.
Works in real-time without disrupting existing workflows.

→ Economic Feasibility

- Open-source + CPU-based system keeps costs low.
- Can scale without expensive GPU infrastructure.

→ User Feasibility

- Automatic detection with minimal user effort.
- **Simple, multilingual alerts** help non-technical users.

→ Technical Feasibility

- With **Digital Signal Processor** + Whisper/Librosa detection engine
- **3 access paths:** Default Dialer , Telecom Integration , VoIP/In-App
- API-first, CPU-based & cloud/edge deployable

VIABILITY

→ Technical & Operational Viability

- Detection system can **adapt to new voice-cloning techniques.**
- Flexible deployment across **Device, Telecom & VoIP.**

→ Economic Viability

- **Multiple revenue streams:** Banks, telecom operators & consumer app.
- Low operating cost with **scalable deployment.**

→ Social & Legal Viability

- **Privacy & consent-first** approach.
- Helps protect **elderly and vulnerable users** from voice scams.

→ Scalability

- One core engine can support **multiple platforms.**
- Can expand from **pilot** → **banks** → **telecom** → **large-scale deployment.**

HOW WE ACCESS CALL AUDIO : PLATFORM FEASIBILITY



Path A: Default Dialer App

Android InCallService gives real call-audio access when VaaniShield is set as the phone app.



Path B: Telecom-Side Integration

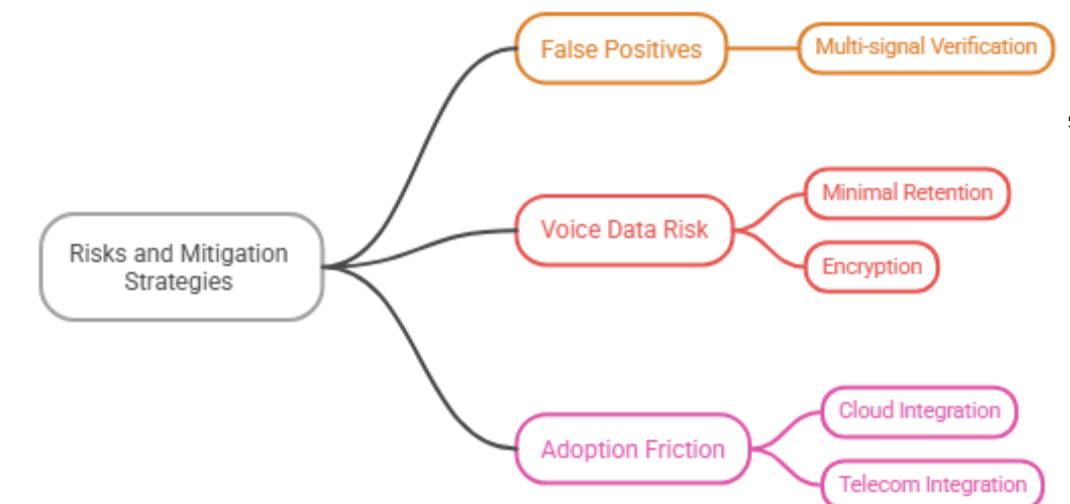
Carriers integrate our detection API at the network level - most scalable, no install needed.



Path C: VoIP / In-App Calling

Direct audio access with no OS permission battle - used for this hackathon's live demo.

POTENTIAL RISK AND MITIGATION STRAGIES



Impacts

Reduced Financial
Fraud

Stronger Cyber
Resilience

Safer Digital
Payments

Protection from
Vulnerable Users

IMPACT

REDUCED FINANCIAL FRAUD

Prevents or interrupts voice-based scams before financial loss occurs.

STRONGER CYBER RESILIENCE

Adds a proactive defense layer against AI-generated voice and social-engineering attacks.

SAFER DIGITAL PAYMENTS

Builds greater confidence in phone-based banking and digital transactions.

PROTECTION FROM VULNERABLE USERS

Provides an additional safety layer for senior citizens and less digitally experienced users.

From Detection to Prevention

Shifts fraud protection from “*detect after the incident*” to “*intervene before the damage.*”

Benefits

Real-Time
Protection

Explainable Alerts

Multilingual Security

Privacy-First

Fraud Prevention

BENIFITS

REAL TIME PROTECTION

Scam calls can be flagged within seconds, before the victim transfers money.

EXPLAINABLE ALERTS

Users understand *why* a call is risky instead of receiving only a generic risk score.

MULTILINGUAL SECURITY

Supports Hindi, Marathi and English, making protection accessible to a wider population.

PRIVACY - FIRST

Designed around minimal audio retention and on-device/edge processing where feasible.

Fraud Prevention

Helps users pause, verify and avoid potentially fraudulent transactions.

AI / DEEPPAKE RESEARCH

AASIST – Audio Anti-Spoofing
Audio deepfake/spoof detection using spectro-temporal analysis.
Source: AASIST research + official implementation.
[click here](#)

RawNet2 – End-to-End Anti-Spoofing
Raw waveform-based approach for detecting spoofed/synthetic speech.
Source: ICASSP 2021 / EURECOM.
[click here](#)

SUPPORTING TECHNOLOGIES

Audio Processing

- MFCC
- CQT Spectrogram
- Raw Waveform

Speech & Context

- Whisper → Speech-to-text
- NLP → Urgency/fraud keywords

Primary Research

47% of Indians exposed to AI voice scams - McAfee

₹1,935 Cr lost to digital-arrest scams in 2025 - MHA / I4C

\$12.5B global contact-center fraud losses in 2024 — Pindrop

DATASET & MODEL RESEARCH

ASVspoof 2019 / 2021:
Benchmark dataset for detecting synthetic speech and training models (like RawNet2).

ECAPA-TDNN:
Architecture used for speaker verification and identity comparison.

Features	Our Website	Traditional Solutions	Other Tools (Generic AI)
Consumer Protection	✓	Limited	✗
Hindi + English + Marathi	✓	✗	English Only
"Why Flagged?" Explainability	✓	✗	✗
Voice + Context + NLP Fusion	✓	✗	✗
Timestamp-Level Detection	✓	✗	✗
Telephony + VoIP + IVR Integration	✓	Limited	✗